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# Does Transparency Hinder Technological Novelty? Evidence from Large Pharmaceutical Firms

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**Abstract.** *Problem definition:* The drug development process has historically lacked transparency, leading to selective reporting of clinical trial results. Although clinical trial transparency aims to address this issue, increased public scrutiny may make managers more risk averse, potentially stifling novel research and development (R&D) initiatives. This raises the following question. Does transparency hinder technologically novel R&D? *Methodology/results:* Using data from nearly 10,000 clinical trials from 2000 to 2014, we examine how a 2005 policy to increase transparency affects drug novelty. Using a continuous difference-in-differences approach, we find that transparency prompts firms to rely more on previously used technologies (i.e., exploitative R&D) without reducing their pursuit of never-before-tested technologies (i.e., exploratory R&D). We identify negative informational spillovers as a mechanism; technology failures that are now more visible because of transparency could erode confidence in related trials, prompting firms toward familiar technologies to avoid reputational damage. Consistent with this mechanism, the impact of transparency is strongest among firms with less-diversified portfolios that are the most exposed to negative informational spillovers. *Managerial implications:* Our primary theoretical contribution is in demonstrating the unintended negative impact of transparency on novelty. For managers, we highlight diversification as a mitigation strategy. We urge policymakers to create incentives to support novel R&D despite transparency pressures.

**Supplemental Material:** The online appendix is available at <https://doi.org/10.1287/msom.2023.0527>.

**Keywords:** pharmaceuticals • clinical trials • new technologies • transparency • natural experiments

## 1. Introduction

Pharmaceutical drug development has historically lacked transparency (Miller et al. 2017). Although firms are required to conduct clinical trials to evaluate the safety and efficacy of drug candidates, they often avoid disclosing clinical trials that produce lackluster results, such as inadequate safety or drug efficacy concerns. This selective reporting, often termed as the “file drawer” problem (Rosenthal 1979), can create an overly positive impression of a drug’s safety and effectiveness. This issue has prompted a series of policies that aim to promote clinical trial transparency.

However, when managerial decisions are subject to public scrutiny because of increased transparency, managers can become less inclined to take risks (Kahneman and Lovaglio 1993) and may favor safer projects to avoid potential public failures. This reduced willingness to take risks could threaten novel research and development (R&D) initiatives given their higher inherent risk, raising the following important question. Does transparency hinder technologically novel R&D?

We study this research question in the context of pharmaceutical drug development for two key reasons. First, developing novel drugs is crucial to a firm’s performance as they could generate blockbuster sales. Additionally, they significantly enhance patient care, underscoring their economic and societal importance. Second, the long timelines and high failure rates in drug development drive risk aversion among firms, leading to underinvestment in innovative R&D and prompting firms to focus on “producing variations of top-selling drugs already on the market” (Angell 2010). The lack of novelty is often attributed to firms’ avoidance of additional risks inherent with pursuing truly innovative therapies (Munos and Chin 2011). These factors provide a compelling case for using the pharmaceutical context to study the impact of transparency on the novelty of drug innovation.

Our analysis takes advantage of the first and evidently most important policy aimed at increasing clinical trial transparency: the 2005 International Committee of Medical Journal Editors (ICMJE) policy. The policy

was designed to increase transparency by requiring firms to disclose trial information through trial registration as a prerequisite for publication in academic journals (De Angelis et al. 2004). The policy was widely adopted by medical journals and led to a sharp increase in clinical trial registration (see Section 3.4 for details), providing a unique opportunity to empirically examine the impact of increased transparency on technologically novel R&D.

We investigate the policy's impact using Cortellis, a data set that tracks clinical trials. Our sample encompasses the drug development efforts of the largest pharmaceutical firms over a 15-year period (2000–2014). To measure a drug's technological novelty, we construct a score based on its pharmacological approach (i.e., the underlying technology defined as the number of times that a given technology has been previously used in clinical trials). The more frequently a technology has been deployed in previously tested trials, the less novel it is considered. We refer to this measure as "*Prior Deployments*," where lower values indicate higher technological novelty. At the extreme, a value of zero denotes a maximally novel technology—one that has never before been tested in clinical development.

Our identification strategy is based on the fact that the ICMJE policy did not affect all firms equally as pharmaceutical firms differ substantially and persistently in their engagement with academic publishing (Arora et al. 2018). We reason that firms with limited prepolicy publishing were less exposed to and less affected by the ICMJE policy than those with higher involvement in publishing. We incorporate this firm-level heterogeneity in publication into a continuous difference-in-differences (DiD) specification.

Our findings indicate that increased transparency introduced by the ICMJE policy steers firms to rely more on technologies that they have used in the past. Specifically, we find that a one-standard-deviation increase in policy exposure is associated with an 11.7% increase in the average number of prior deployments, indicating a shift toward less novel and less risky R&D. This result is robust to falsification tests and alternative measures of policy exposure and novelty.

One potential mechanism behind this transparency-driven shift toward less novel projects is the risk of negative informational spillovers. In drug development, such spillovers occur when the failure of one technology leads stakeholders to draw pessimistic conclusions about the viability of other trials based on the same technology (Magazzini et al. 2012, Krieger 2021). Because transparency makes failures more difficult to hide, firms may gravitate toward less novel (and inherently less risky) technologies to minimize the negative consequences of public failures. Therefore, we expect firms with less diversified technology portfolios to be particularly vulnerable to these spillovers

as each failure affects a larger share of their pipeline. Indeed, we find that the negative effect of transparency on novelty is concentrated among firms with relatively low technological diversification.

Further corroborating the mechanism, we find that the effect is stronger for drugs developed in house compared with those acquired externally. It is likely that internally developed technologies originate in a firm's core capabilities; therefore, failures in these projects are more likely to trigger negative inferences about related technologies in the firm's pipeline. In contrast, externally acquired technologies are less likely to be integrated with the firm's own technology base and may, therefore, carry lower informational spillover risk.

An important implication of these findings is that increased transparency may inadvertently deter firms from pursuing never-before-tested frontier technologies. We explore this concern in our post hoc analysis and ask the following question. Does increased transparency lead firms to reduce the pursuit of novel technologies? The answer to this question entails substantial societal implications because of the importance of medical innovation as well as strategic consequences for the firms, whose long-term success often depends on rare but radical breakthrough drugs.

To investigate this, we distinguish between exploratory R&D (which involves technologies never tested before) and exploitative R&D (based on previously used technologies). Our analysis suggests that transparency does not affect exploratory R&D (i.e., firms continue to invest in projects involving never-before-tested technologies). In contrast, firms increase their exploitative R&D efforts, where firms increase their use of established technologies with a history of prior use. Organizational ambidexterity or a firm's simultaneous pursuit of exploration and exploitation explains this finding (March 1991, Swift 2016). Firms respond to transparency by reshaping their portfolios—increasing low-risk exploitative efforts while preserving high-risk exploratory efforts.

To summarize, transparency prompts firms to lean more heavily on previously used technologies—an effect that is most pronounced in firms with low technological diversification and in-house developed drugs. Although exploratory R&D remains unaffected, firms increase exploitative R&D efforts, suggesting a strategic rebalancing of portfolios toward less novel technologies. By demonstrating the previously undocumented trade-off between transparency and novelty, our results make strong contributions to the existing literature. Our results are especially relevant in the context of drug development, where transparency is critical for safeguarding patients' well-being but implies long-term costs in terms of less innovative drugs reaching the market. Our study has important implications for policymakers, too, who may need to design transparency

mechanisms that encourage accountability without discouraging breakthrough innovation in the pharmaceutical sector.

The remainder of the paper is structured as follows. First, we provide a brief review of the relevant literature streams, the contextual details surrounding drug development, and the ICMJE policy. Next, we describe the data, the operationalization of our key variables, and the research method. We then describe our main results and provide evidence of the mechanism to corroborate these results. We conclude by discussing the contributions of our study to management theory, policymakers, and practice.

## 2. Literature Review and Hypotheses

### 2.1. Literature Review

Our study is closely associated with two literature streams: transparency and innovation. Below, we review the transparency literature and focus on the new product development (NPD) context within the innovation stream.

**2.1.1. Transparency.** Transparency has multidisciplinary roots (for a comprehensive review, see Bernstein 2017). We summarize the most pertinent studies in Table A.1 in the Online Appendix and highlight several interesting characteristics. We find that the unit of analysis in a vast majority of studies is an individual, such as a customer (Mohan et al. 2020), an employee (Beer et al. 2021), or a physician (Chao and Larkin 2022). We also note that the majority of studies consider “voluntary” transparency initiatives, such as promoting social responsibility initiatives (Kraft et al. 2018) and firm authenticity (Mejia et al. 2019). In general, we find that increased transparency leads to positive, desirable outcomes, such as higher productivity (Mas and Moretti 2009), higher service quality (Buell and Norton 2011), and higher customer purchase intentions (Buell and Kalkanci 2021).

Our study differs from existing research in several important ways. To the best of our knowledge, it is the first empirical study to examine how NPD transparency affects the novelty of new products. A noteworthy exception is Brown and Martinsson (2019), who explored the consequences of transparency on the *amount of innovation* at a national level. In contrast, our study focuses on understanding the firm-level repercussions of heightened transparency on the *novelty of innovation*. Moreover, although most studies examine the effect of transparency on individuals in a service setting, our study is conducted at the firm level in the NPD context. Finally, our focus is on examining the potential undesirable effects of transparency, whereas others investigate its desirable effects (Lee et al. 2021).

**2.1.2. Innovation.** In the NPD innovation literature, the notions of novelty and risk are interrelated such that the level of risk is thought to vary with the “extent of the novelty.” To capture the varying levels of risk, researchers have drawn a distinction between radical and incremental innovation (e.g., Peng et al. 2008, Hoang and Rothaermel 2010). Radical innovation implies exploration that is higher risk but provides firms with a first-mover advantage, leading to superior performance (Kirchhoff 1992, Chandy and Tellis 2000). In contrast, incremental innovation is associated with low-risk exploitation and allows a firm to leverage its existing knowledge and technologies and build on available firm-specific resources, resulting in more cost-efficient product development (March 1991).

Although radical and incremental innovation can be viewed as a trade-off, where more exploitation implies less exploration and vice versa, prior research has demonstrated that most successful firms undertake both exploratory and exploitative activities: an approach popularized as ambidexterity (Swift 2016). Ambidexterity balances high-risk, high-reward exploratory initiatives with low-risk, low-reward exploitative endeavors, ensuring that exploiting existing knowledge does not crowd out novel, exploratory initiatives (Benner and Tushman 2003). We contribute to the ambidexterity literature by providing evidence that firms respond ambidextrously to the transparency shock.

### 2.2. Hypotheses Development

**2.2.1. The Impact of Transparency on Novelty.** Managers often prefer less risky options over more risky ones to avoid potential losses, even if the latter options are associated with larger expected payoffs (Tversky and Kahneman 1992). This is particularly true in the NPD context, where choice sets are often large, the likelihood of failure is high, and the risk-reward payoffs are uncertain (Kahneman and Lovallo 1993). In such contexts, decision makers can become more risk averse, especially if they are likely to be held accountable for their choice (Viscusi et al. 1987). Consequently, when managerial decisions are open to public view, the fear of potential failure could drive managers to choose safer options.

Specifically, increased transparency could lead to a diminished “tolerance for failure,” defined as the ability or willingness to endure negative project outcomes (Manso 2011), an effect that is particularly salient in NPD contexts (Hutchison-Krupat and Chao 2014). This greater reluctance toward risk-taking could reduce managerial effort toward novel NPD initiatives because lower risk-taking is synonymous with lower novelty (Peng et al. 2008, Dranove et al. 2022). To reduce the likelihood of failure, firms may choose to exploit previously used technologies that leverage existing knowledge over exploring novel technologies that provide a

competitive advantage and superior financial performance but are more prone to failure. Therefore, we hypothesize the following.

**Hypothesis 1.** *Transparency is negatively associated with technological novelty.*

### 2.2.2. Moderating Role of Technological Diversification.

Transparency may compel firms to be more risk averse and avoid potential failures. This phenomenon is particularly pronounced in drug development, where failures are common, with 9 in 10 drugs failing to make it to market (Wong et al. 2019). Such failures not only adversely affect firm valuation and performance (Giotra et al. 2007, Hermosilla 2021) but also create negative spillovers for other trials (i.e., when a drug fails in a trial, stakeholders grow pessimistic about other drugs utilizing the same technology) (Magazzini et al. 2012, Krieger 2021).

We propose that firms developing portfolios of closely related technologies may be more susceptible to negative informational spillovers created by enhanced transparency than firms with unrelated technology portfolios. For example, a hypothetical Firm A, which has invested in multiple trials relying on a specific technology, may consider discontinuing these projects following disappointing trial results because of concerns about the technology's viability. In contrast, a hypothetical Firm B with a more technologically diversified portfolio facing a similar trial failure will likely be better insulated from these adverse effects. Hence, we hypothesize the following.

**Hypothesis 2.** *Technological diversification moderates the relationship between transparency and technological novelty such that the negative effect of transparency on novelty is more pronounced for firms with lower technological diversification.*

## 3. Research Context

### 3.1. The Development Process for New Pharmaceutical Drugs

The drug approval process in the United States requires detailed evidence demonstrating the drug candidate's safety and efficacy. This evidence is produced through clinical trials, in which the drug candidate's effects are evaluated on humans. The clinical trial process is highly structured, comprising three successive phases. The first of these, Phase 1, corresponds to brief, small-scale studies on healthy volunteers that typically enroll 10–20 patients and last one to two months. These studies aim to ensure that the candidate possesses basic properties for its safe evaluation in larger populations. If these properties are established, subsequent Phase 2 trials are executed as a proof of concept for the conjectured therapeutic benefits (efficacy). When the presence of such

benefits is supported, a final testing round, Phase 3, is carried out. Phase 3 trials seek to provide definitive proof of the drug's therapeutic merits. Accordingly, these trials are the longest (several months to years) and the largest (several hundred to thousands of patients) in the tiered process. Once trials conclude, sponsors submit data to the Food and Drug Administration (FDA), which evaluates the drug's market suitability and provides usage guidelines.

An important aspect of this process is that preclinical and clinical trial data are legally considered trade secrets at the federal and state levels (Kesselheim and Mello 2007), providing sponsors a basis for secrecy. Sponsors are not required to report all investigational activities to the FDA. When sponsors choose to submit a subset of these activities, the FDA's ability to disclose this information is limited by these trade secret protections.

### 3.2. Incentives Opposing Transparency

Clinical trials generate information that can be highly sensitive to the sponsor's operations. Particularly, negative clinical trial outcomes, such as poor efficacy or safety concerns, create strong incentives for sponsors to limit transparency. Unfavorable results can lead to negative perceptions of the product's quality, reducing consumer demand. For example, Le Noury et al. (2015) found that GlaxoSmithKline (GSK) withheld studies that raised concerns about the efficacy of the drug Paxil in treating depression. Similarly, in the cases of the painkiller Vioxx and the antidiabetic Avandia, their respective manufacturers, Merck and GSK, withheld information pertaining to the drugs' safety profiles. In all of these cases, judicial or regulatory proceedings established that information had been withheld in an attempt to minimize negative impacts on commercial outcomes.

Lackluster clinical trial results may also lead to negative inferences regarding the potential of other firms' products, particularly those relying on the same technology. This type of spillover is illustrated by the publication of a study that debunked the SMARCA4 ligand as a target for lung cancer (Vangamudi et al. 2015). Commenting on the study, a renowned industry observer wrote that "a number of programs (in academia and industry both) came to a juddering, dust-spewing halt once people saw the data" (Lowe 2015). Prior research presents evidence that negative clinical trial results from one firm may prompt other firms to terminate related trials (Krieger 2021) and can reduce the subsequent citations received from other firms (Magazzini et al. 2012).

### 3.3. Distorted Clinical Evidence and the Push for Transparency

The high-profile cases cited above bolstered the view that the industry suffered from a significant selective

reporting, also known as the “file drawer” problem (Rosenthal 1979). This problem can endanger patients because of the distortion of the available evidence for clinical decision making (Greenberg 1999).

Paxil, widely used in treating major depressive disorders in pediatric populations, became the poster child. In 2003, a few years after the drug’s developer, GSK, was granted FDA approval, Paxil was linked to increased risks of teenage suicide. In March 2004, an internal company memo dated 1998 first surfaced to reveal that the company had suppressed data suggesting Paxil’s lack of efficacy (Kondro 2004). Shortly thereafter, *The Lancet* published a scientific review concluding that the suppressed data indicated that the drug’s associated risks in pediatric populations were greater than its benefits (Whittington et al. 2004). By June 2004, the New York State Attorney General sued GSK (Sataline 2005), and the UK regulator opened its own investigation (Durman and Rushe 2004). These events precipitated the decision of medical journal editors to announce a protransparency policy later that same year.

3.4. The ICMJE Transparency Mandate

Responding to the public outcry caused by the Paxil saga, the ICMJE, a consortium of editors from the 12 most influential medical journals, instituted a clinical trial transparency policy in September 2004 through an editorial simultaneously published in the 12 journals (De Angelis et al. 2004).

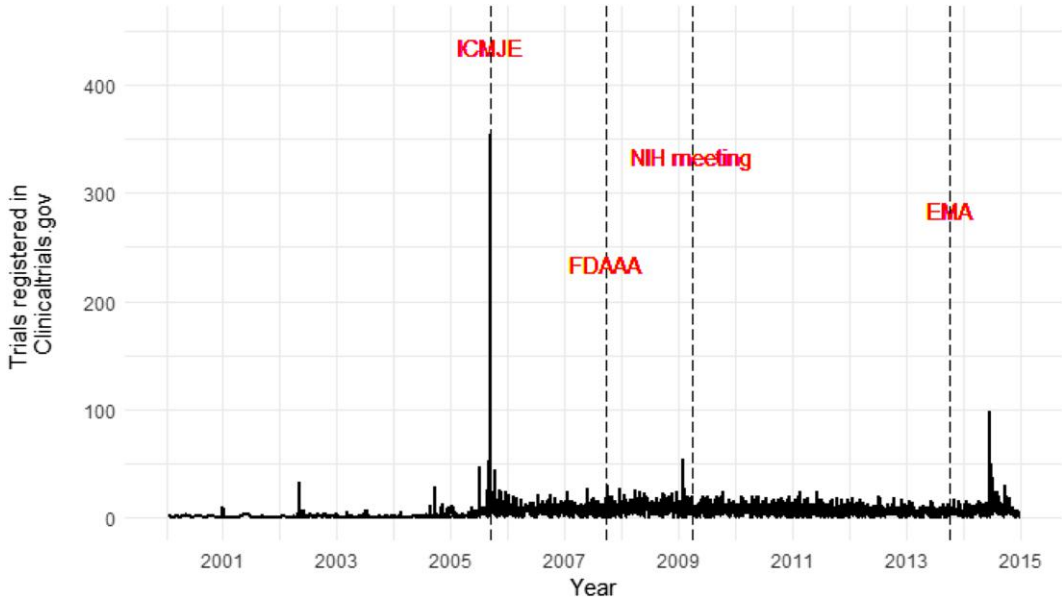
The ICMJE mandate targeted the widely prevalent practice of academic publishing by scientists working

in pharmaceutical R&D (see Section 4.4 for a discussion of the importance of academic publishing in the pharmaceutical industry). The new rule required the timely registration of clinical trials as a precondition for considering research for publication in the journals. It mandated that trials starting after July 1, 2005 would have to be registered before the first patient was enrolled, and for trials already underway, retrospective registration should occur no later than September 13, 2005. Registration consisted of submitting trial information (trial design, intervention, patient groups, outcomes, etc.) to a publicly accessible clinical trial database or “registry.” This was expected to discourage the cherry-picking of results that reach the public domain, thus providing healthcare providers and patients a more balanced view of drugs’ risks and benefits (Brownlee 2004).

Medical journals swiftly adopted the policy, triggering a sustained rise in trial registrations. We illustrate this by plotting the number of trials registered on ClinicalTrials.gov (CTGOV) in Figure 1. Compared with an average of 3 trials per day in the prepolicy period, the number of registered trials per day jumped to its all-time highest number of 354 trials on the policy’s cutoff date of September 13, 2005. Zarin et al. (2005), who examined CTGOV registrations between May and October of 2005, found that the total number of studies registered on the site had grown by 73% over the six-month period.

Beyond the one-time change, patterns delineated in Figure 1 also suggest that the ICMJE policy led to an

Figure 1. (Color online) Count of Clinical Trials Registered in ClinicalTrials.gov by Year



Notes. The vertical dashed lines show the dates of various policies that could impact trial registrations. The increase in the number of registered clinical trials was significantly more following the ICMJE policy when compared with any other policy implemented during the study duration. EMA, European Medicines Agency; FDAAA, Food and Drug Administration Amendments Act; NIH, National Institute of Health.

enduring change toward greater transparency. The postpolicy trial registration rate settled at eight trials per day, an approximate threefold increase compared with the average rate of three trials per day before the launch of the ICMJE policy.

### 3.5. Subsequent Policies

Several other policies targeting enhanced clinical trial transparency were subsequently enacted. Some of the more noteworthy policies include the 2007 Food and Drug Administration Amendments Act, the 2009 National Institute of Health public meeting, and the 2013 European Medicines Agency press release. However, the ICMJE policy is considered the most important and the most consequential effort toward protransparency. This is evident in Figure 1, which reveals that the increase in the number of trials registered following the ICMJE policy eclipsed the increase following other similar policies by a wide margin.

Experts also acknowledge the pioneering role of the ICMJE policy. According to Zarin et al. (2017, p. 388), the ICMJE policy “instigated a cascade of events that have ... transformed the trial reporting system.” Evidence from our discussions with R&D portfolio managers in the pharmaceutical industry also substantiated the pre-eminent nature of the ICMJE policy. The head of clinical research at an innovative global biopharmaceutical firm referred to the ICMJE policy as the “tsunami of change” for pharmaceutical firms at that time. In our empirical analysis, we employ the ICMJE policy as the exogenous shock, which enhanced industry-wide transparency levels by requiring that all clinical trials be registered to be considered for academic publication.

## 4. Methods

### 4.1. Research Design

We use the enactment of the ICMJE policy to investigate how enhanced transparency levels affect firms’ willingness to pursue novel drug R&D. We do not assume that the policy completely resolved the industry’s secrecy issues. Instead, we view it as introducing an industry-wide boost in transparency that increased the cost of withholding trials from the public domain by requiring trial registration in order to publish in academic journals.

We derive identification from the differential impact of the ICMJE policy across firms. Because academic publishing is more important to some firms compared with others (see Section 4.4 for more details), we argue that a firm with a higher publication intensity before the policy would be more heavily affected by the policy when compared with a firm with a lower publication intensity. Our econometric framework incorporates this variation through a prepolicy “exposure” variable, which tracks publication intensity defined as each

firm’s publishing relative to the number of drug development projects before the ICMJE policy. We combine this cross-sectional prepolicy exposure variation with the timing of the policy event to create the continuous DiD model.

The continuous DiD approach is increasingly being used by researchers in various contexts (e.g., Parker et al. 2016, Yurukoglu et al. 2017, Alpert et al. 2018, Dranove et al. 2022). Similar to our study, these studies also use preperiod data to construct their own “policy exposure” measure. For example, Parker et al. (2016) evaluates the impact of a text message ban on price dispersion. Their “exposure” measures the heterogeneity in preban usage of the information platform, arguing that the impact of the ban would be greater for units that used the platform relatively more. Similarly, to estimate how the 2006 insurance expansion of Medicare Part D impacted drug innovation, Dranove et al. (2022) computes “exposure” as the share of prescriptions issued to Medicare enrollees in the preperiod disease categories. Likewise, we compute our exposure measure as each firm’s publication intensity in the preperiod (see Section 4.4).

### 4.2. Drug Pipeline Data

We rely on drug development data compiled by Clarivate Analytics’ Cortellis, which is a pharmaceutical intelligence database that tracks worldwide drug development activities. It collects information on new clinical trials from a variety of sources (e.g., clinical trial registries from around the world, company reports and websites, academic articles and conferences, regulatory agencies, and specialized media among others). The data are collected and certified by over 500 experts and are widely recognized as the industry’s most comprehensive and up-to-date source. Cortellis has recently been used in a number of research articles (Subramanian et al. 2020, Hermosilla 2021, Krieger 2021, Dranove et al. 2022).

As part of the data certification process, Cortellis’ experts link each trial with a series of variables. These include the tested drug and sponsoring firm, the trial phase, and the initiation date. Crucial to our study, linked variables also include information about the drug’s underlying pharmacological approach, which we simply refer to as “technology.” We use this information to construct a measure of novelty.

Employing the Cortellis data, we construct a data set where each observation is a clinical trial. Because the stated effects of transparency require sufficiently large portfolios of drugs, we focus on a sample of trials sponsored by large pharmaceutical firms. We do so by focusing on firms that are members of PhRMA, the consortium of the largest developers in the pharmaceutical industry. Of the 33 firms that were active PhRMA members during 2021, we were able to incorporate 22

into our analysis given the data limitations. Eleven firms were excluded for the following reasons. (i) Two firms were established around or after the policy change, (ii) two firms had little to no drug development prior to the policy change, (iii) six firms did not have publication data, and (iv) one privately held firm was excluded because of the lack of control variables. For the included firms, we harmonized firm names. For example, trials linked to Sanofi include trials that Cortellis links to the sponsors “Sanofi Canada,” “Sanofi GmbH,” “Sanofi Inc.,” etc. We also link trials sponsored by subsidiaries and joint ventures. For example, trials sponsored by the Sanofi and Bristol-Myers Squibb (BMS) joint venture are assigned to both Sanofi and BMS.

Our data distinguish drugs that were developed internally and acquired externally through mergers, acquisitions (data for Pfizer include the drug “ABX-0401,” which it acquired because of its acquisition of Wyeth), or licensing (e.g., data for Pfizer also include the drug “PF-06647263,” which it licensed from Stemcentrx, a start-up focused on developing anticancer drugs). Consistent with prior literature, 40% of the drug candidates in the sample were acquired externally (Achilladelis and Antonakis 2001).

We compiled data for trials initiated between 2000 and 2014. This period yields a relatively long “postpolicy” period (about nine years), which is advantageous in light of the relatively slow pace of drug development. Our final sample consists of 9,353 clinical trials. Figure A.1 in the Online Appendix shows the distribution of trials across therapeutic areas and phases. Oncology is the largest therapeutic area, accounting for 18.2% of trials, followed by neurology (11.5%) and immunology (8.6%). Phase 1, Phase 2, and Phase 3 trials account for 29.0%, 43.7%, and 27.3% of trials, respectively.

A potential concern in our analysis is whether the policy might have affected the observability of the Cortellis data. Specifically, could the ICMJE policy have also impacted Cortellis’ ability to gather data? This inability to gather data would be evident as discontinuities around the policy implementation period. However, an important factor mitigating this concern is that Cortellis allows for backfilling information from late disclosures. In other words, if a trial initiated at date  $t_0$  is made public at date  $t_1 > t_0$ , the trial will be included in the data set at  $t_1$  while associated with the  $t_0$  start date. To empirically assess this concern, we plot the time series of trial initiations in the Cortellis data set (see Figure A.2 in the Online Appendix) and find continuous growth throughout the sample, without any visible discontinuities around the policy implementation period.

### 4.3. Technological Novelty

Researchers have used many approaches to measure technological novelty (for a review, see Kesselheim

et al. 2013). The majority of these measures are based on approved drugs, such as generic versus innovator drugs, the share of drugs receiving regulatory designations, or “first in class” drugs. One fundamental drawback of such measures is that they overlook drugs that are never approved, which represent the majority of pharmaceutical R&D.

To rectify this and conceptualize a metric that (i) focuses on scientific novelty and (ii) can score the novelty of candidates that never reach the market, we rely on the novelty measure proposed by Dranove et al. (2022) and later utilized by Arslan et al. (2024). This measure is based on the descriptions of the technology employed by each drug (“target-based action”), which are culled from the Cortellis database. These descriptions systematically codify the pharmacological approach employed by the drug tested in each clinical trial. Each technological description contains two pieces of information: (i) the targeted biological entity and (ii) the mechanism used to target it. For example, the technology described as “5-HT2 receptor antagonist” acts by antagonizing (i.e., blocking the response of) 5-HT2 receptor proteins. Similarly, “5 $\alpha$ -reductase inhibitor” describes a drug that acts by inhibiting the function of 5 $\alpha$ -reductase enzymes. This approach to pharmacological classification spans a broad technological space: almost 8,000 unique technologies in the full Cortellis data set going as far back as the mid-1970s (1,307 covered by our data duration).

We label this novelty score as “Prior Deployments.” For trial  $j$  testing drug  $d$ , we define prior deployments as the number of other drugs previously tested in clinical trials that rely on the same technology as drug  $d$ . Each deployment will produce information about the technology, which makes each incremental use less novel. Therefore, we associate smaller values of prior deployments with higher levels of novelty. To illustrate the construction of the variable, consider the following two examples. When AstraZeneca initiated a Phase 3 trial for its drug Quetiapine, the drug’s associated technology (5-HT2 receptor antagonist) had been previously used by 48 other drugs tested in clinical trials. Accordingly, we associate this trial with *Prior Deployments* = 48. In turn, when GlaxoSmithKline started a Phase 2 trial for its drug Izonsteride in 2001, the associated technology (5 $\alpha$ -reductase inhibitor) had previously been used by 11 other drugs tested in clinical trials (*Prior Deployments* = 11). Because the latter trial tested a drug with fewer prior deployments, we can infer it to be a more novel development than the former. Within the full sample, *Prior Deployments* have an average of 16.3, indicating that an average technology is used approximately 16 times. Row (6) in Table 1 shows that this value varies significantly across the trials sponsored by different firms in the sample.

**Table 1.** Descriptive Statistics

| Variables                  | Time period | Minimum | Maximum | Mean  | Median | Standard deviation |
|----------------------------|-------------|---------|---------|-------|--------|--------------------|
| (1) Number of trials       | 2000–2014   | 87.0    | 1,205.0 | 422.7 | 300.0  | 330.5              |
| (2) Number of publications | 2000–2005   | 11.0    | 1,118.0 | 309.1 | 292.0  | 293.0              |
| (3) Number of drugs        | 2000–2005   | 3.0     | 185.0   | 42.8  | 24.0   | 47.0               |
| (4) Raw exposure           | 2000–2005   | 0.8     | 38.4    | 8.9   | 8.7    | 8.0                |
| (5) Exposure               | 2000–2005   | −1.0    | 3.7     | 0.0   | 0.0    | 1.0                |
| (6) Technological novelty  | 2000–2014   | 0.0     | 124.0   | 16.3  | 10.0   | 18.8               |

Notes. Descriptive statistics for all variables are reported at a firm level except for technological novelty, which is reported at a trial level. The period 2000–2005 corresponds to the period from January 1, 2000 to September 13, 2005, which is the prepolicy period. The period 2000–2014 corresponds to the period from January 1, 2000 to December 31, 2014, which is the total study duration.

*Prior Deployments* may vary by phase (early stage versus late stage) and type of trial (exploration versus exploitation). We provide a time variation plot for early-stage trials (Phases 1 and 2) and late-stage trials (Phase 3) using dashed and solid lines, respectively, in panel (a) of Figure A.3 in the Online Appendix. Both series exhibit a positive trend. That is, the number of times that a technology is deployed increased over time, implying that trials became less novel over time. This supports claims of insufficient investment in novel technologies (Angell 2010, Munos and Chin 2011). We also observe that late-stage trials are associated with higher *Prior Deployments* compared with early-stage trials, suggesting that less novel projects are more likely to advance farther in the process, plausibly because of their lower risk. In our econometric analyses, we account for the differences over time and across clinical trial phases by including year and phase fixed effects. We also plot the proportion of *exploratory trials* (i.e., trials associated with technologies that have zero prior deployments) (see panel (b) of Figure A.3 in the Online Appendix) and find a fairly stable temporal rate of approximately 8%. We further investigate this variation when attempting to decompose the main effects of transparency on novelty in Section 6.3.

#### 4.4. Policy Exposure

The ICMJE mandate sought to increase clinical trial transparency by requiring trial registration as a prerequisite for journal publication. Therefore, firms that were more involved in academic publishing should have been more heavily affected by the policy. Varying publication intensities across firms could be explained by a variety of organizational practices, such as firm-specific evaluation of the costs and benefits of publishing (Arora et al. 2018). Publishing intensities may be lower if firms invest less in internal scientific research, face a higher risk of informational spillovers to rival firms, or fear that publications will thwart their ability to patent. On the flip side, publishing may offer reputational advantages (Polidoro and Theeke 2012), strengthen demand for the firm's products (Azoulay 2002), facilitate

collaboration with biotechnology firms, and strengthens ties with the scientific community (Lerner and Malmen-dier 2010). It has also been suggested that some firms may use publishing as a perk to attract scientists (e.g., Zucker and Darby 1997).

Based on these observations, we construct a policy exposure variable that quantifies the importance of academic publishing for each firm. This variable reflects the organizational strategy that is hard to change. This interpretation is supported by the results discussed in Section 5.4.3, which show that the ICMJE policy did not fundamentally impact firms' publication behavior. For each firm  $i$ , we define the policy exposure variable as

$$Exposure_i = \text{StandardNormal}\left(\frac{Publications_i}{Drugs_i}\right). \quad (1)$$

The numerator (*Publications*) corresponds to the number of articles published in academic journals that reported funding from firm  $i$ . We compute this using the Altmetric database by counting all articles funded by the firm. In cases where an article has multiple funding sources, we count the article under each of the associated organizations. We compute the denominator (*Drugs*) using the Cortellis data by counting the number of new drugs that the firm introduces for clinical trials. To avoid contamination with the policy's effects, both variables (numerator and denominator) are measured using data from before the policy's creation. Over the prepolicy period, firms in our sample were associated with a total of 7,571 academic publications and introduced 808 drugs prior to the policy's enactment (see rows (1)–(3) in Table 1 for summary statistics). For ease of interpretation of our estimates, we standardize the publications-to-drugs ratio such that the exposure variable has a mean of zero and a standard deviation of one. Positive values indicate above-average publication intensities, whereas negative values indicate below-average intensities.

Given the formulation of Equation (1), we assign larger policy exposure to firms that published more articles per drug. For firms with high exposure, we expect that failure to comply with the policy will result in considerable disruptions to their intellectual

endeavors. By contrast, the policy’s enactment may not have impacted firms with low exposure. Hence, we expect that the firms with high publication intensities will have more prominently adjusted their innovation decisions in response to the ICMJE policy. Raw and normalized exposure values are shown in rows (4) and (5) in Table 1, respectively. As anticipated by prior literature, the exposure values demonstrate significant heterogeneity across firms.

## 5. Results

In this section, we explain our findings by first presenting model-free evidence, undertaking a formal analysis, offering an intuitive interpretation of the estimated coefficients, and finally, conducting a series of robustness checks on our results.

### 5.1. Model-Free Evidence

We illustrate the idea that firms may change their technological choices in response to enhanced transparency in Table 2. We first separate trials (in columns) based on whether their sponsor firm’s policy exposure is below or above the median. We further separate trials based on whether they were initiated before or after the policy implementation on September 13, 2005. Each cell of Table 2 shows the average prior deployments of trials. We note that above-median exposure firms exhibited a disproportionate increase in prior deployments following the policy’s enactment compared with below-median exposure firms. The pattern holds separately for all trials (row (A) in Table 2), early-stage trials (row (B) in Table 2), and late-stage trials (row (C) in Table 2). For instance, in row (A) in Table 2, the average prior deployment increases by 4.51 for firms with below-median exposure (column (4) in Table 2) and by 6.23 for firms with above-median exposure (column (7) in Table 2), indicating that both groups shifted to less novel technologies after the transparency shock. Moreover, the average prior deployments for firms with above-median policy exposure (6.23) are higher compared with firms with below-median policy exposure (4.51), resulting in a positive difference of 1.72 (column (8) in Table 2). These

patterns support the hypothesis that firms with greater exposure to the policy disproportionately reduced the novelty of their R&D because of the enhanced transparency regime.

### 5.2. Econometric Estimates

To investigate whether the conclusions in the prior section formally hold, we estimate the following continuous DiD specification employing the *Exposure* measure from Equation (1). The model uses prior deployments as the dependent variable:

$$\begin{aligned} \mathbb{E}[\text{Prior Deployments}_j] \\ = f(\beta \cdot \text{Exposure}_{i(j)} \times \text{Post}_j + \Theta \cdot X_j + \Delta \cdot Z_{i(j), \text{year}(j)} \\ + \lambda_{i(j)} + \tau_{\text{year}(j)} + \mu_{\text{phase}(j)}), \end{aligned} \quad (2)$$

where  $j$  indicates trials and  $i(j)$  indexes trial  $j$  mapped to firm  $i$ . Within our data window that extends from January 1, 2000 to December 31, 2014, the variable *Post* is an indicator for trials initiated after the policy’s implementation (i.e.,  $\text{Post} = \mathbb{1}[\text{trial } i \text{ initiated after September 13, 2005}]$ ). Given that *Prior Deployments* is a count variable, we estimate the equation with Poisson quasimaximum likelihood regression, which accounts for overdispersion. We cluster standard errors at the firm level.

Equation (2) includes a series of controls. The vector  $X$  contains therapeutic area dummy variables and an indicator for whether the tested candidate is a biological or chemically synthesized agent. In addition,  $Z$  represents financial variables collected from Compustat at a firm-year level—R&D intensity operationalized as the ratio of R&D expense over sales, size operationalized as the natural logarithm of total assets and the natural logarithm of the number of employees, and technological diversification operationalized as the natural logarithm of the Herfindahl–Hirschman Index (HHI) based on the share of technologies in a firm’s portfolio. Finally, the specification includes firm, year, and phase specific fixed effects ( $\lambda$ ,  $\tau$ , and  $\mu$ , respectively). Note that the addition of firm and year fixed effects would make redundant the inclusion of the stand-alone *Exposure* and *Post* variables.

**Table 2.** Model-Free DiD Evidence

| Average prior<br>deployments<br>(1) | Below-median exposure |             |                               | Above-median exposure |             |                               | Difference in<br>differences<br>(8) = (7) – (4) |
|-------------------------------------|-----------------------|-------------|-------------------------------|-----------------------|-------------|-------------------------------|-------------------------------------------------|
|                                     | Pre<br>(2)            | Post<br>(3) | Post – pre<br>(4) = (3) – (2) | Pre<br>(5)            | Post<br>(6) | Post – pre<br>(7) = (6) – (5) |                                                 |
| (A) All trials                      | 13.98                 | 18.49       | 4.51                          | 11.90                 | 18.13       | 6.23                          | 1.72                                            |
| (B) Early-stage trials              | 12.94                 | 15.94       | 3.00                          | 11.45                 | 15.90       | 4.45                          | 1.45                                            |
| (C) Late-stage trials               | 18.00                 | 25.03       | 7.03                          | 13.74                 | 25.33       | 11.59                         | 4.56                                            |

*Notes.* Higher values of prior deployments reflect less novel drugs. The preperiod is from January 1, 2000 to September 13, 2005, whereas the postperiod is from September 14, 2005 to December 31, 2014. Early-stage trials belong to either Phase 1 or Phase 2 trials. Late-stage trials are Phase 3 trials. Positive DiD in the last column indicates that firms that are more exposed to the policy disproportionately increase prior deployments more than firms that are less exposed to the policy.

The parameter of main interest in Equation (2) is  $\beta$  (associated with *Exposure*  $\times$  *Post*). A  $\hat{\beta} > 0$  would point to a disproportionate increase of *Prior Deployments* among trials sponsored by more exposed firms, supporting the hypothesis that transparency pushes firms to increase their reliance on used technologies.

We present parameter estimates in Table 3. Column (1) in Table 3 shows the results obtained from a regression without the control variables. Consistent with the model-free evidence in Section 5.1, the continuous DiD effect estimate is positive and statistically significant, thus providing support for Hypothesis 1. The 0.18 estimate associates one-standard-deviation-higher policy exposure with an average 19.7% increase in prior deployments following the policy's enactment. Columns (2)–(7) in Table 3 present estimates of models that incrementally add fixed effects and controls to the base model. Across these specifications, the continuous DiD estimate of interest is positive and statistically significant. In the full model of column (7) in Table 3, the estimate  $\hat{\beta} = 0.11$  ( $p < 0.05$ ) associates one-standard-deviation-higher policy exposure with an 11.7% increase in the expected value of *Prior Deployments*.

### 5.3. Novelty Effects as Technological Lag

A common challenge faced by empirical frameworks aimed at quantifying drug novelty is that the proxies that they rely upon operate on self-contained scales (e.g., Dranove et al. 2022, Krieger et al. 2022). This feature makes it difficult to interpret estimates. What does the estimated 11.7% increase in *Prior Deployments* mean in practice? To address this question, we consider the accumulation of deployments by a technology given its repeated use in other drug development

processes. This approach allows us to compute the time that an average technology would take to accumulate 11.7% additional deployments.

In panel (c) of Figure A.3 in the Online Appendix, semitransparent curves show the logarithm of the number of deployments that each technology has accumulated  $N$  years after its first deployment (horizontal axis). The solid curve in panel (c) of Figure A.3 in the Online Appendix represents the average pattern. Given the log transformation, the linear pattern indicates that technologies accumulate deployments at a roughly constant rate of 6.3% additional deployments per year. This implies that an 11.7% increase can be interpreted as equivalent to falling behind the technological frontier by almost 1.9 years.

### 5.4. Robustness Checks

**5.4.1. Pretrends and Temporal Unfolding.** We investigate the effect's temporal unfolding and the presence of confounding pretrends using an event study version of Equation (2), where we interact *Exposure* with year-specific indicators. The estimated parameters are presented in Figure 2, with the effect for 2005 normalized to zero. Parameter estimates for the years 2000–2004 exhibit a flat trend and are statistically insignificant at conventional levels under both individual and joint significance tests ( $p = 0.74$ ). This pattern does not support the presence of confounding pretrends. After the policy's enactment, the continuous DiD parameters adopt a gently increasing trend and are mostly statistically significant with 95% confidence. The estimate for 2014, the last year covered by the sample, shows that one-standard-deviation-higher policy exposure is associated with approximately 55.8% higher expected

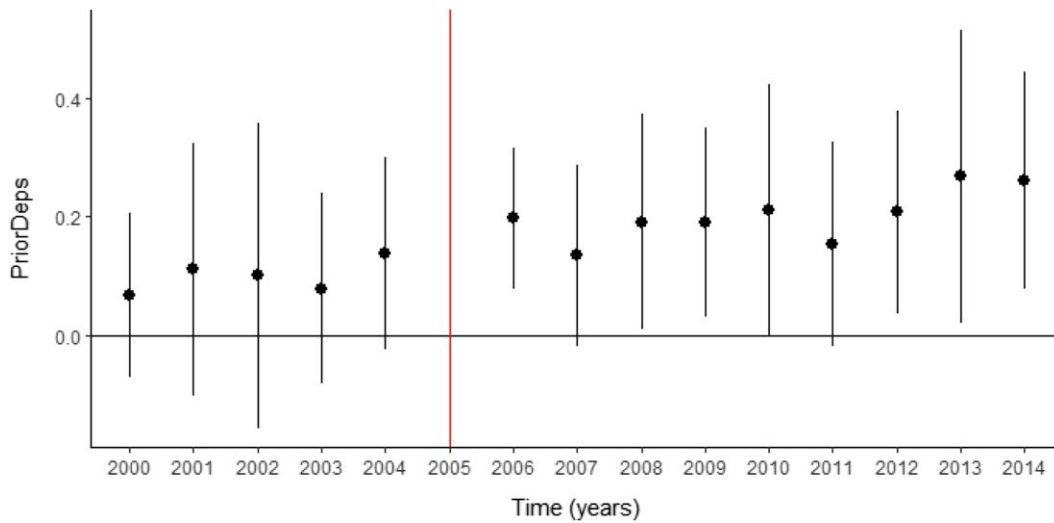
**Table 3.** Main Effect—Poisson Quasimaximum Likelihood Continuous DiD Estimation

| Dependent variable: <i>Prior Deployments</i> | (1)               | (2)               | (3)              | (4)             | (5)             | (6)              | (7)              |
|----------------------------------------------|-------------------|-------------------|------------------|-----------------|-----------------|------------------|------------------|
| <i>Post</i>                                  | 0.27***<br>(0.06) | 0.27***<br>(0.06) |                  |                 |                 |                  |                  |
| <i>Exposure</i>                              | −0.04<br>(0.05)   |                   |                  |                 |                 |                  |                  |
| <i>Exposure</i> $\times$ <i>Post</i>         | 0.18**<br>(0.08)  | 0.14**<br>(0.06)  | 0.12**<br>(0.06) | 0.11*<br>(0.06) | 0.12*<br>(0.07) | 0.13**<br>(0.06) | 0.11**<br>(0.05) |
| $N$                                          | 9,353             | 9,353             | 9,353            | 9,353           | 9,353           | 9,353            | 9,353            |
| Firm FE                                      | No                | Yes               | Yes              | Yes             | Yes             | Yes              | Yes              |
| Year FE                                      | No                | No                | Yes              | Yes             | Yes             | Yes              | Yes              |
| Therapeutic area FE                          | No                | No                | No               | Yes             | Yes             | Yes              | Yes              |
| Biologic FE                                  | No                | No                | No               | No              | Yes             | Yes              | Yes              |
| Phase FE                                     | No                | No                | No               | No              | No              | Yes              | Yes              |
| Compustat controls                           | No                | No                | No               | No              | No              | No               | Yes              |

*Notes.* Robust standard errors (clustered at the firm level) are presented in parentheses. We use Poisson quasimaximum likelihood estimation because our dependent variable is an overdispersed count variable. We incrementally add control variables to the base model presented in column (1). Adding firm fixed effects (FEs) in column (2) omits the estimation of the coefficient associated with the *Exposure* variable. Similarly, adding year FEs in column (3) omits the estimation of the coefficient associated with the *Post* variable.

\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

**Figure 2.** (Color online) Event Study Specification: Continuous DiD Effects over Time



Notes. This figure reports the event study version of Table 3. Each marker represents the coefficient of the interaction between the *Exposure* variable and an indicator variable for each year of the study duration. The interaction coefficients are calculated relative to those of the year 2005. The bars represent 95% confidence intervals.

**Prior Deployments.** These results suggest that the policy's effects on technological novelty may persist and continue to intensify over time.

To further investigate the presence of pretrends, we estimate the following specification only using data from before the policy's creation:

$$\begin{aligned} \mathbb{E}[\text{Prior Deployments}_j] \\ = f(\delta \cdot \text{Exposure}_{i(j)} \times \text{Time}_j + \Theta X_j + \lambda_{i(j)} \\ + \tau_{\text{year}(j)} + \mu_{\text{phase}(j)}), \end{aligned} \quad (3)$$

where  $\text{Time}_j$  represents the time elapsed since the start of our study duration (i.e., January 1, 2000). A confounding pretrend would be supported by an estimate  $\hat{\delta} > 0$  as this would indicate that our main documented effects were brewing before the policy's creation. Instead, the estimated coefficient shown in Table A.2 in the Online Appendix is insignificant. Together, these results suggest that there is no evidence of pretrends.

**5.4.2. Falsification.** To further probe the causal interpretation of our main estimate of column (7) in Table 3, we replace observed firm-level *Exposure* scores with random values drawn from a standard normal distribution. Figure A.4 in the Online Appendix shows the distribution of continuous DiD estimates ( $\hat{\beta}$ ) obtained from estimating Equation (2) in 1,000 pseudosamples created following this procedure. The distribution of estimates is centered at zero and is symmetric. As shown by the vertical dashed line in Figure A.4 in the Online Appendix, our main estimate from column (7) in Table 3 is located in the right tail of the distribution,

yielding an implied  $p$ -value of 0.018. In other words, the estimate is statistically improbable relative to the distribution of estimates obtained via randomization. The falsification results lend confidence to the causal interpretation of our main results.

**5.4.3. Pre- and Postpolicy Publishing Behavior.** Recall that the ICMJE policy used publication practice as a means to promote trial transparency. One may wonder whether firms change their publication behavior in response to the ICMJE policy and whether such changes drive our results. Our analysis presented in Online Appendix A.1 provides convincing evidence that the ICMJE policy did not systematically impact publication behavior.

**5.4.4. Alternate Novelty Proxies.** Our novelty measure, *Prior Deployments*, counts all prior drugs entering the clinical trial process that have used a given technology. To test the robustness of this measure, we rerun our analysis with other operationalizations of the metric. We leverage the idea that some “related” or “validated” technologies might be less risky than others and test whether firms increase their usage following the transparency policy. In Online Appendix A.2, we discuss these operationalizations in greater detail and show that our results are robust to alternate measures of novelty.

## 5.5. Early- vs. Late-Stage Trials

Here, we consider whether transparency differentially affects the development of novel technologies in early-stage (Phases 1 and 2) versus late-stage (Phase 3) trials. This analysis is motivated by the observation

that late-stage trials receive greater attention from investors and the media compared with early-stage trials. Thus, lackluster late-stage results will cause firms greater distress. Hence, we expect that the effects of transparency will be stronger in late-stage trials than in early-stage trials. To examine this, we split trials by stages into early-stage (Phases 1 and 2) trials and late-stage (Phase 3) trials, and we estimate Equation (2) separately in the two subsamples. Consistent with our prediction, column (1) in Table A.5 in the Online Appendix shows that transparency did not have a significant effect on early-stage trials. In contrast, the estimate obtained from late-stage trials in column (2) in Table A.5 in the Online Appendix shows that the impact of transparency is statistically significant and primarily driven by this subsample.

## 6. Mechanism and Implications

In this section, we explore the heterogeneity of our effect across types of firms, innovation sources, and types of technologies.

### 6.1. Technological Diversification as a Moderator of Informational Spillovers

We first explore which firms are more impacted by transparency. We argue that under a regime of heightened transparency, where information about disappointing test results is more difficult to conceal, firms that are more vulnerable to negative informational spillovers may disproportionately reduce investments in technologically novel drug development projects. In particular, we posit that firms whose portfolios consist of highly related technologies may be more vulnerable to negative spillover risk compared with firms developing portfolios of unrelated technologies. Accordingly, we proxy for each firm's vulnerability to negative spillovers

through a measure of technological diversification. Following Garcia-Vega (2006), we compute an HHI for firm-level technological concentration as

$$\text{Technological diversification}_i = \sum_{t \in T_j} \left( \frac{|C_t|}{|\mathcal{P}_j|} \right)^2, \quad (4)$$

where for trial  $j$  sponsored by firm  $i$ , we consider all other drugs being tested in clinical trials by the same firm at the time that the trial begins. We call this set  $\mathcal{P}_j$ . Next, we identify the set of all technologies associated with these candidates,  $T_j$ , and the set of drugs associated with each technology  $t$ ,  $\{C_t\}_{t \in T_j}$ . The resulting measure ranges from zero (full technological diversification) to one (full technological concentration). The sample average (2.26) and the standard deviation (4.07) suggest that firms in our sample are generally diversified, although with substantive variation. In addition, we test whether firms altered their levels of technological diversification in response to the new policy by estimating Equation (2) but with technological diversification as our dependent variable. The coefficients from Table A.6 in the Online Appendix show that the ICMJE policy did not affect diversification.

Next, we categorize our data using the median value of HHI into two subsamples: below-median HHI and above-median HHI, which correspond to high and low levels of technological diversification, respectively. The results are presented in Table 4. For trials sponsored by firms with relatively high technological diversification (below-median HHI), our estimate reveals a statistically insignificant effect of enhanced transparency (column (1) in Table 4). By contrast, for trials sponsored by technologically concentrated firms, the effect is positive and statistically significant. Moreover, its magnitude is

**Table 4.** Diversification as a Moderator—Poisson Quasimaximum Likelihood Continuous DiD Estimation

| Dependent variable: <i>Prior Deployments</i>        | (1)                                                         | (2)                                                        | (3)              |
|-----------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------|------------------|
| <i>Exposure</i> × <i>Post</i>                       | −0.10<br>(0.09)                                             | 0.17***<br>(0.03)                                          |                  |
| <i>Exposure</i> × <i>Post</i> × <i>log(TechDiv)</i> |                                                             |                                                            | 0.06**<br>(0.03) |
| Sample                                              | Below-median<br>HHI = high technological<br>diversification | Above-median<br>HHI = low technological<br>diversification | All              |
| <i>N</i>                                            | 4,861                                                       | 4,492                                                      | 9,353            |
| Firm FE                                             | Yes                                                         | Yes                                                        | Yes              |
| Year FE                                             | Yes                                                         | Yes                                                        | Yes              |
| Therapeutic area FE                                 | Yes                                                         | Yes                                                        | Yes              |
| Biologic FE                                         | Yes                                                         | Yes                                                        | Yes              |
| Phase FE                                            | Yes                                                         | Yes                                                        | Yes              |
| Compustat controls                                  | Yes                                                         | Yes                                                        | Yes              |

Notes. Robust standard errors (clustered at the firm level) are presented in parentheses. We use Poisson quasimaximum likelihood estimation because our dependent variable is an overdispersed count variable. FE, fixed effect.

\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

about 50% larger than the baseline estimate from the full sample (column (7) in Table 3). As an additional check, we assess the moderating role of diversification by including a triple-interaction term. The estimated coefficient (column (3) in Table 4) is positive and statistically significant, indicating that more heavily technologically concentrated firms may have reacted more strongly to the policy because of more pervasive negative informational spillovers, providing support for Hypothesis 2.

6.2. Internal vs. External Innovation

Large pharmaceutical firms often acquire innovation externally from other firms (Achilladelis and Antonakis 2001), especially radical innovation (Comanor and Scherer 2013). The distinction between internal innovation and external innovation raises an important question. How does transparency influence firms’ external and internal innovation strategies? Building on the evidence for the spillover mechanism presented in the previous section, we argue that failures in internally developed projects, rooted in a firm’s core technological capabilities, are more likely to trigger negative inferences about other technologies in the pipeline. By contrast, externally acquired compounds, obtained through licensing, mergers, or acquisitions, may entail lower informational spillover risks.

To investigate this, we classify drug candidates as either internally developed or externally acquired using our data (see Section 4.2 for more details). We then re-estimate our main econometric specification (Equation (2)) separately for the two subsamples. The results (Table 5) show that the coefficient for internal innovation is positive and statistically significant (column (1) in Table 5), whereas the coefficient for external innovation is not (column (2) in Table 5). This suggests that

transparency increases firms’ reliance on internal innovation, consistent with the idea that firms respond to higher reputational risk by adjusting strategies more strongly in areas where technological spillovers are more pronounced.

6.3. Does Enhanced Transparency Thwart R&D Exploration?

Our central result points to a discouraging trade-off; enhanced transparency in drug development may imply a reduction in technological novelty because of greater exposure of R&D failure risks. In this section, we aim to further qualify this relationship by examining the impact of transparency on the exploitation-exploration spectrum (Chandrasekaran et al. 2015), where novelty could be considered a matter of degrees (Boudreau et al. 2016). Specifically, we ask the following question. Do firms shy away from groundbreaking (exploratory) R&D, or do they invest more in follow-on, incremental (exploitative) research activities?

A focus on reduced exploration would suggest that enhanced transparency may jeopardize the future development of lifesaving novel drugs. Given that the development of novel drugs is already at alarmingly low levels (Angell 2010, Munos and Chin 2011), a further reduction should be of great concern to policymakers. Moreover, novel breakthroughs are critical for organizations too as they serve as a means to develop future knowledge for sustained competitive advantage and long-term firm growth and survival (Kirchhoff 1992). Therefore, a better understanding of which types of technologies on the exploitation-exploration spectrum are impacted by increased transparency should be of interest to managers and policymakers alike.

To investigate this empirically, we define a trial as purely novel if it has zero prior deployments using a binary variable that equals one when *Prior Deployments* = 0. We reanalyze Equation (2) using a logit model with the new indicator variable as our dependent variable. The results (column (1) in Table 6) show a statistically insignificant coefficient, suggesting that enhanced transparency levels may not thwart exploratory R&D.

Next, to investigate the effects of transparency on exploitative R&D, we estimate the likelihood of a firm pursuing technologies that might not be considered purely novel. To distinguish between a technology that has only been used a few times (and hence, has a higher degree of risk and novelty) from a technology that has been extensively used, we classify previously used technologies into four categories using quartile values as the cutoffs. Further, we create four indicator variables that each take a value of one if a trial belongs to the quartile and zero otherwise. For instance, the dependent variable associated with the first quartile takes the value of one for trials that are the closest to

Table 5. Poisson Quasimaximum Likelihood Continuous DiD Estimation on Subsamples by Innovation Source

| Dependent variable: Prior Deployments | Internal innovation (1) | External innovation (2) |
|---------------------------------------|-------------------------|-------------------------|
| <i>Exposure × Post</i>                | 0.14***<br>(0.04)       | −0.04<br>(0.09)         |
| <i>N</i>                              | 5,488                   | 3,865                   |
| Firm FE                               | Yes                     | Yes                     |
| Year FE                               | Yes                     | Yes                     |
| Therapeutic area FE                   | Yes                     | Yes                     |
| Biologic FE                           | Yes                     | Yes                     |
| Phase FE                              | Yes                     | Yes                     |
| Compustat controls                    | Yes                     | Yes                     |

Notes. Robust standard errors (clustered at the firm level) are presented in parentheses. We use Poisson quasimaximum likelihood estimation because our dependent variable is an overdispersed count variable. FE, fixed effect.  
\**p* < 0.1; \*\**p* < 0.05; \*\*\**p* < 0.01.

**Table 6.** Logistics Continuous DiD Estimation with Modified *Prior Deployments* as Dependent Variables

|                        | Exploration    | Exploitation    |                |                    |                   |
|------------------------|----------------|-----------------|----------------|--------------------|-------------------|
|                        | First use      | Minimal use     | Moderate use   | Frequent use       | Extensive use     |
|                        | (1)            | (2)             | (3)            | (4)                | (5)               |
| <i>Exposure × Post</i> | 0.14<br>(0.14) | −0.15<br>(0.13) | 0.11<br>(0.17) | −0.37***<br>(0.14) | 0.32***<br>(0.08) |
| <i>N</i>               | 9,353          | 9,353           | 9,353          | 9,353              | 9,353             |
| Firm FE                | Yes            | Yes             | Yes            | Yes                | Yes               |
| Year FE                | Yes            | Yes             | Yes            | Yes                | Yes               |
| Therapeutic area FE    | Yes            | Yes             | Yes            | Yes                | Yes               |
| Biologic FE            | Yes            | Yes             | Yes            | Yes                | Yes               |
| Phase FE               | Yes            | Yes             | Yes            | Yes                | Yes               |
| Compustat controls     | Yes            | Yes             | Yes            | Yes                | Yes               |

*Notes.* Robust standard errors (clustered at the firm level) are presented in parentheses. We use logistics estimation because of the binary dependent variables. The dependent variable in column (1) is coded one if the technology is used for the first time and zero otherwise. The dependent variables in columns (2)–(5) are coded one if *Prior Deployments* of a technology is in the first, second, third, or fourth quartile of *Prior Deployments*, respectively. FE, fixed effect.  
\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

purely novel technologies in the set. We label each of these categories minimal use, moderate use, frequent use, and extensive use, corresponding to quartiles 1, 2, 3, and 4, respectively.

We then estimate Equation (2) using a logit model with four binary dependent variables. Results presented in columns (2)–(5) in Table 6 show that transparency impacts firms’ R&D decisions only for the “frequent use” and “extensive use” technologies. The coefficient is negative and statistically significant for “frequent use” technologies (column (4) in Table 6), indicating that firms with higher exposure to the policy are less likely to use a frequently used technology. However, the coefficient for “extensive use” technologies (column (5) in Table 6) is positive and statistically significant, implying a greater proclivity to pursue the least novel technologies.

Together, these results suggest that firms increase their use of existing technologies without reducing their exploration of novel ones. Interestingly, they appear more cautious toward moderately novel technologies that have been frequently used in the past. This pattern reflects an ambidextrous strategy, where firms simultaneously conduct both exploitative and exploratory activities (He and Wong 2004). This ambidextrous approach may be an effort to balance high-risk exploratory R&D, despite transparency pressures, in anticipation of future gains, with exploitative efforts aimed at near-term gains. Perhaps firms pursue ambidexterity because failures in truly novel drugs are expected and thus, less damaging, whereas failures in less novel drugs send a stronger negative signal.

Finally, we assess how transparency affects firms’ strategies across four categories: internal exploration, internal exploitation, external exploration, and external exploitation. To construct these subsamples, we combine the internal-external distinction introduced in

Section 6.2 with the exploration-exploitation classification. Specifically, we identify whether each drug candidate represents an exploratory or exploitative effort and whether it originated internally or was acquired externally. We then re-estimate our main econometric specification (Equation (2)) separately for each of the four resulting subsamples. We find that the estimated coefficient for internal exploitation (column (3) in Table 7) is positive and statistically significant, indicating that firms react to increased transparency with a greater focus on internal exploitation. None of the other categories show statistically significant effects. As a set, these results reaffirm our main conclusion while providing greater granularity; transparency encourages increased exploitation, particularly through internal innovation activities.

### 7. Discussion

Clinical trial transparency is vital for preserving the integrity of the medical literature and patient welfare. However, our findings indicate that greater transparency can inadvertently steer firms toward using less novel technologies. Although we find no evidence that transparency suppresses purely exploratory R&D, regulators must remain vigilant to its unintended consequences as disclosure requirements become more stringent. To counteract the potential risks, transparency initiatives should be paired with targeted incentives that preserve firms’ appetite for high-risk, high-reward R&D. For instance, regulators may consider policies that extend market exclusivity for novel drugs or bolster tax credits akin to those under the FDA’s Orphan Drug Designation. They could also expand expedited pathways, such as the Breakthrough Therapy Designation, to provide additional regulatory support for truly pioneering candidates. Both types of incentives have proven highly effective in encouraging innovative R&D (Yin 2008, Hermosilla 2022).

**Table 7.** Continuous DiD Estimation on Subsamples by Innovation Source and Exploration/Exploitation

|                        | Internal exploration<br>(1)      | External exploration<br>(2)      | Internal exploitation<br>(3)                 | External exploitation<br>(4)                 |
|------------------------|----------------------------------|----------------------------------|----------------------------------------------|----------------------------------------------|
| <i>Exposure × Post</i> | −0.18<br>(0.42)                  | 0.37<br>(12.02)                  | 0.14***<br>(0.04)                            | 0.001<br>(0.08)                              |
| Dependent variable     | $\mathbb{1}[\text{PriorDeps}=0]$ | $\mathbb{1}[\text{PriorDeps}=0]$ | <i>PriorDeps</i>                             | <i>PriorDeps</i>                             |
| Sample                 | Internal innovation              | External innovation              | Internal innovation and <i>PriorDeps</i> > 0 | External innovation and <i>PriorDeps</i> > 0 |
| N                      | 5,488                            | 3,865                            | 5,064                                        | 3,547                                        |
| Model                  | Logit                            | Logit                            | PQML                                         | PQML                                         |
| Firm FE                | Yes                              | Yes                              | Yes                                          | Yes                                          |
| Year FE                | Yes                              | Yes                              | Yes                                          | Yes                                          |
| Therapeutic area FE    | Yes                              | Yes                              | Yes                                          | Yes                                          |
| Biologic FE            | Yes                              | Yes                              | Yes                                          | Yes                                          |
| Phase FE               | Yes                              | Yes                              | Yes                                          | Yes                                          |
| Compustat controls     | Yes                              | Yes                              | Yes                                          | Yes                                          |

Notes. Robust standard errors (clustered at the firm level) are presented in parentheses. We use logistics estimation when the dependent variable is binary and Poisson quasimaximum likelihood (PQML) estimation when the dependent variable is an overdispersed count variable. FE, fixed effect.  
\* $p < 0.1$ ; \*\* $p < 0.05$ ; \*\*\* $p < 0.01$ .

From a macro perspective, transparency-driven risk aversion may help explain the continued growth of outsourced pharmaceutical R&D. Estimates suggest that more than half of new drugs are brought to market through collaborations between biotech start-ups and more established pharmaceutical firms (Kneller 2010). Collaborating with academic research centers and specialized start-up firms to perform early-stage research allows large pharmaceutical firms to transfer early risks in the drug discovery process to their partners. It also provides them with the opportunity to acquire/in license the less risky, late-stage drug development from those partners (Angell 2021). Anecdotal evidence suggests that large pharmaceutical firms have leveraged such partnership models to avoid patent cliffs in the past (Gibney 2022). Transparency may also have contributed to the growth of outsourcing high-risk, early-stage drug discovery because it allows firms to avoid the potential negative informational spillovers arising from increased transparency.

Our finding that transparency mainly affects firms with concentrated portfolios suggests a potential strategy for managers to mitigate this effect. Specifically, technological diversification could curb negative informational spillovers and reduce the higher failure costs associated with increased transparency. Although specialization has been highlighted as a key strength that leads to better learning and productivity (Narayanan et al. 2009), firms with high dependence on a small set of technologies may be more vulnerable to environmental shocks. Our study adds to the existing literature that highlights the benefit of diversification in innovation management (Klingebiel and Rammer 2014).

The moderating effect of diversification on the impact of transparency is particularly relevant in our context because pharmaceutical firms face increasing pressure

to be transparent. Many firms have publicly committed to “open innovation” and pledged to disclose information about clinical trials and their results (Hunter and Stephens 2010). At the same time, there is a notable trend of novel drugs being discovered by firms with more specialized portfolios. The increasing need for transparency, coupled with the high reliance on R&D originating in firms with specialized technological portfolios, bolsters the importance of diversification in limiting the negative impact of transparency on technological novelty.

8. Conclusion and Limitations

Our study examines how a transparency shock, the 2005 ICMJE clinical trial registration policy, affects technological novelty in drug development. Using a continuous difference-in-differences design, we find that heightened transparency induces an 11.7% increase in the use of previously deployed technologies—roughly a two-year lag behind the technological frontier—while leaving exploratory, never-before-tested R&D unchanged. We posit that by exposing trial failures and amplifying negative informational spillovers, transparency raises the reputational cost of high-risk projects, driving firms to favor proven technologies. Consistent with this mechanism, only firms with less-diversified technology portfolios shift toward exploitative R&D; those with broader portfolios remain unaffected.

One important caveat regarding our analysis is that it is circumscribed to a segment of the drug development sector: large pharmaceutical firms. Although we expect that transparency may also affect the innovation decisions of smaller firms (e.g., biotech start-ups), we suspect that the mechanisms at play may be different. Perhaps most prominently, these differences may

stem from the size of technological choice sets. Because large firms have broad portfolios and the ability to in license, they have ample scope to choose “the right technology” to develop. In contrast, a typical biotech start-up firm is centered around developing a specific technology and may lack the technological choices that large firms have. Accordingly, if transparency affects innovation by small firms, it may do so via entry/exit rather than through portfolio composition. Financing structures (public investments for large firms and venture capital for small ones) and the role of academic publishing introduce further differences that we believe may be important. From a practical standpoint, the ICMJE policy may be inadequate to study how smaller biotech firms react to increased transparency initiatives, especially given their high turnover rates. A few smaller firms may have survived the entire study duration. Perhaps, future researchers could examine the impact of transparency initiatives that are more appropriate and relevant to small firms.

Although our study highlights an unintended negative effect of increased transparency on technological novelty, we consider it important to acknowledge the benefits of transparency for the medical enterprise. The ICMJE policy was implemented to address problems associated with the selective reporting of clinical trial results in the pharmaceutical industry and may have led to benefits for key stakeholders, including patients, firms, and providers. For instance, the increase in trial registration following the ICMJE policy may have instilled greater confidence among trial subjects, resulting in an increase in trial participation. For pharmaceutical firms, increased transparency might have deterred firms from hiding or misreporting trial outcomes (such as with Paxil), leading to a reduction in the number of corporate violations and misdemeanors. By highlighting trial results that demonstrate the poor safety or efficacy of trial drugs, the policy may have helped physicians make more informed decisions when prescribing drugs. Future researchers could quantify the benefits of transparency by examining these relationships. Additionally, researchers could also examine the negative consequences of transparency on other measures to help generalize our findings.

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